

Extended Gorin Model for Radical–Radical Recombination Reactions

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Radical–radical recombination reactions (e.g., $\text{CH}_3 + \text{CH}_3 \rightleftharpoons \text{C}_2\text{H}_6$) proceed with no barrier through simple-fission transition states. The application of transition state theory (TST) to these reactions is discussed, achieving a new understanding of the dividing surface and dynamical assumption implicit in all TST treatments of these reactions. A reinterpretation of the modified Gorin model for such transition states is discussed which removes several inconsistencies from this model and greatly improves data prediction and interpretation for radical–radical recombination reactions (and the reverse unimolecular dissociations) in the gas phase. The suggested model is an extension of the basic Gorin approach, which treats the transition state as consisting of two moieties which have the same vibrational and rotational properties as the fully separated fragments. The method discussed here proposes an improvement of the modified Gorin model Hamiltonian that better describes simple-fission reaction dynamics by completely excluding trajectories occurring with unfavorable orientations of the combining moieties from the transition state theory rate coefficient. This new approach is sufficiently simple that the description is applicable to any system and thus can be routinely implemented with modest computational resources. Comparison with experiment and with more precise theoretical descriptions for ethane and neopentane decomposition reactions shows that this treatment provides quantitative agreement for ethane. It is also concluded that more sophisticated treatments of transitional modes than afforded by hindered rotor models are needed for the description of transition states with bulky moieties at elevated temperatures, such as the neopentane decomposition system described here.

I. Introduction

The reactant in a simple-fission reaction is typically a closed-shell species that dissociates to two free radicals; the reverse recombination usually has no barrier to reaction. The development of models for the calculation of unimolecular and recombination rate coefficients for simple-fission reactions in the gas phase is desirable because of the need to know such rate coefficients for modeling and interpretation of combustion processes, atmospheric chemistry, plasma etching, etc. These processes generally involve many reactions whose experimental rate coefficients are either unknown or can be obtained only in the pressure dependent regime; prediction or extrapolation of these quantities is thus vital to technological and environmental progress. The methods of choice for routine calculations of rate coefficients for such reactions are RRKM theory and transition state theory (TST) for microcanonical and canonical descriptions, respectively.

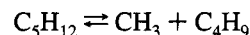
This paper provides a description of the transition state theory rate coefficient of reactions proceeding through a simple-fission transition state which retains much of the simplicity and qualitatively correct physical description of a modified Gorin treatment (see below), while providing a major extension of this model which encompasses qualitative and quantitative improvements on its fundamental precepts.

The application of transition state theory to simple-fission reactions is described in section II, and an evaluation of the modified Gorin model provided in section III; more detail can be found elsewhere.^{1,2} Section IV addresses the principal faults of the modified Gorin model with the derivation of a new description that is implemented and evaluated in section V for

two much discussed examples of simple-fission reactions: (i) ethane decomposition,



and (ii) the decomposition of neopentane,



II. Transition State Theory

RRKM theory provides a microcanonical statistical description of the reaction rate coefficient; TST is its canonical counterpart. These theories rely on the statistical assumption of rapid randomization of energy in the reactant configuration and a dynamical approximation, the transition state assumption, whereby all trajectories that pass through a dividing surface toward products are assumed to have reacted. Variational transition state theory (at the microcanonical or canonical level) recognizes that the transition state assumption implies that the calculated rate coefficient is necessarily an upper bound to the exact rate coefficient; thus, the optimal dividing surface is that which minimizes the calculated rate coefficient.

Angular momentum conservation is crucial to simple-fission reactions; the description employed here is that often used in TST calculations.¹ The principal effect of angular momentum conservation is the availability of energy present in the system's external inactive rotations for assisting the reacting system over the centrifugal barrier. Consider Figure 1, showing the external rotational axes for the C_2H_6 molecule. Rotational energy in motion around the x and z axes of the molecule is considered to be inactive, i.e. not available for randomization in the equilibrium configuration of ethane. As dissociation proceeds,

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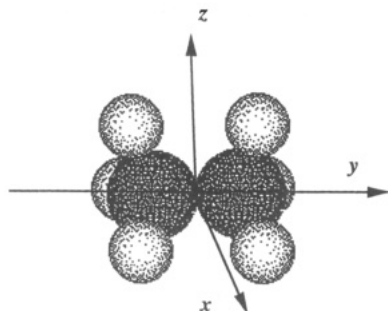


Figure 1. External rotational axes of ethane.

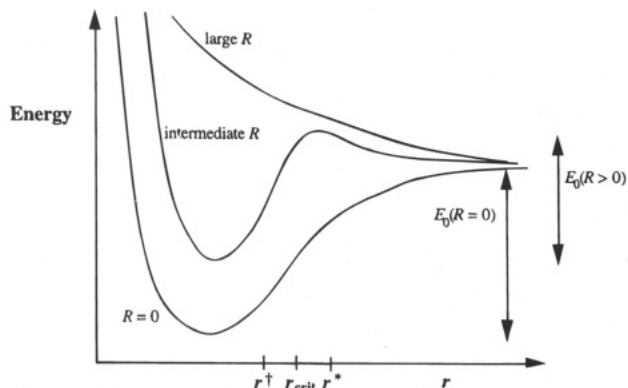


Figure 2. Effective potential for varying rotational energies. Note the decreasing critical energy E_0 for reaction for increasing R . Possible values of the parameters $r^\dagger$, r_{crit} , and r^* are shown; note that $r^\dagger < r_{\text{crit}} < r^*$.

the energy in these rotations is released into the reaction coordinate as angular momentum is conserved.

Rotation about the y axis is considered active, but is assumed to correspond to only a small component of the total angular momentum, thus enabling the external inactive rotational energy to be treated as approximately conserved. This is not strictly true for polyatomic systems, but for systems such as C_2H_6 the inactive rotational energy R shall be assumed² to be approximately given by $BJ(J+1)$, where B is the inactive external rotational constant. Then the active energy E and R (in the dissociating molecule's equilibrium configuration) are independently conserved quantities for the isolated molecule. We have that

$$R = BJ(J+1) \quad (1)$$

As the reaction proceeds, the rotational energy changes as the rotational constant changes, and thus the rotational energy $R^\dagger$ at the transition state is given by

$$R^\dagger = B^\dagger J(J+1) = (B^\dagger/B)R \quad (2)$$

The variation of the rotational energy $R^\dagger$ with the separation of the fragments at the transition state $r^\dagger$ contributes to the effective potential

$$V_{\text{eff}}(r^\dagger) = R^\dagger + V(r^\dagger) \quad (3)$$

where $V(r)$ is the potential function of the two fragments interacting with a radial separation r . The principal features of the effective potential (see Figure 2) are the presence of the centrifugal barrier at $r = r^*$ and the elevated energy of the reacting species, both of which vary in magnitude with increasing R . The critical energy for reaction $E_0(R)$ is given by

$$E_0(R) = V_{\text{eff}}(r^*) - R \quad (4)$$

The effects of angular momentum conservation on the rate coefficient can thus be included (by using the pseudodiatomic approximation) in a TST rate coefficient by accounting for the effects of the effective potential on the critical energy for reaction and on the energy available for reaction. The microscopic rate coefficient $k(E, J)$ is then given by²

$$k(E, J) = \frac{\int_0^{E - V_{\text{eff}}(r^\dagger)} \varrho^\dagger(E_+) dE_+}{hQ(E)}, \quad E > V_{\text{eff}}^* \quad (5)$$

It can be shown² that for a transition state that is independent of E and J (but not necessarily at the top of the centrifugal barrier) the canonical average over eq 5 gives for the high-pressure limit

$$k_{\text{uni}}^\infty = \frac{k_B T}{h} \frac{1}{Q_{\text{rot}} Q_a} \int_0^\infty \frac{dE_{\text{rot}}}{B} (e^{-V_{\text{eff}}^*/k_B T} \int_0^{\Delta V_{\text{eff}}} dE_+ \varrho^\dagger(E_+) + e^{-V_{\text{eff}}^\dagger/k_B T} \int_{\Delta V_{\text{eff}}}^\infty dE_+ \varrho^\dagger(E_+) e^{-E_+/k_B T}) \quad (6)$$

where

$$\Delta V_{\text{eff}} = V_{\text{eff}}^* - V_{\text{eff}}^\dagger \quad (7)$$

In eq 6, $\varrho^\dagger$ is the density of states of active modes, Q_{rot} is the overall external rotational partition function of reactant, and Q_a is the overall internal (vibrations plus internal rotations) partition function of reactant. Equation 6 is frequently used for canonical variational transition state theory,^{3,4} where angular momentum conservation is taken into account but the transition state is chosen to be optimal for a particular temperature, rather than for a fixed energy and angular momentum.

There are a number of sophisticated descriptions of the Hamiltonians⁵⁻⁷ of such reactions; see also the microcanonical variational calculations of Hase and Zhu.⁸ The present work describes an alternative simple model, based on the modified Gorin model, which, although less precise than these more sophisticated treatments, offers a number of advantages. This model accounts for the essential physics of the process, requires only modest computational resources to evaluate, and is readily transferable, by which is meant that the required parameters can be easily obtained for any chosen system. Of course, results as accurate as more sophisticated treatments cannot be obtained and are only of qualitative predictive value for bulk systems, for reasons that are further elucidated in section V.

III. The Modified Gorin Model

The approach discussed here uses the basic ideas of the Gorin approach:⁹ the transition state for a simple-fission reaction $\text{AB} \rightarrow \text{A} + \text{B}$ is assumed to comprise two loosely separated moieties whose Hamiltonian is that of the species A and B, together with terms corresponding to the interaction between these moieties. Thus, the transition state is approximated as that of two moieties, each of which has the vibrational Hamiltonian of the isolated molecular or atomic fragment, and pivots independently about its component atom of the breaking bond. The independent rotors are each taken to be symmetric tops. It is assumed that the reaction coordinate $r^\dagger$ is the length of the breaking bond.

The Gorin Hamiltonian thus assumes a separable kinetic energy and that the intermoiety potential is zero (i.e., that the rotations of the moieties are independent of each other); the dynamics are thus those of the following independent entities: two two-dimensional rotors (one for each moiety), one one-dimensional rotor (the torsional motion of the moieties coun-

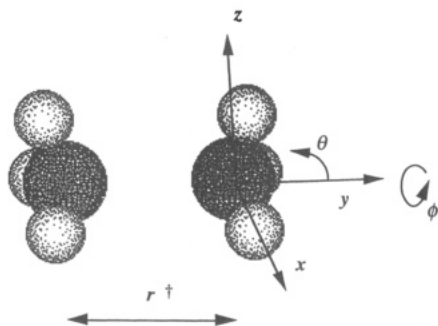


Figure 3. Angles for internal rotation of a methyl fragment. θ corresponds to rotation in the xy plane away from the $\text{CH}_3\text{--CH}_3$ bond axis; θ_{hin} is half the available angle for rotation as hindered by hard-sphere interactions with the other fragment. ϕ corresponds to rotation about the bond axis. r^+ is the fragment separation at the transition state.

terrotating), and the overall rotation of the combined moieties (one two-dimensional and one one-dimensional rotation). The classical contribution to the transition state partition function of these degrees of freedom is then given (e.g., Gilbert and Pitt¹) by

$$Q_{\text{rot}}^{\ddagger} = \pi(k_{\text{B}}T)^4 \prod_{i=1}^2 \frac{1}{\sigma_i} \left(\frac{1}{B_i} \right)^{1/2} \prod_{j=1}^3 \frac{1}{\sigma_j B_j} \quad (8)$$

where the B 's are the rotational constants corresponding to the moments of inertia and the σ_i are the symmetry numbers for each rotation; the B_j in the second product are for rotations about the two (nearly) degenerate axes of the fragment, whilst the B_i in the first product are for rotation about the remaining (single) principal axis. By inverse Laplace transform, the corresponding density of states is

$$\varrho_{\text{rot}}^{\ddagger}(E) = \frac{\pi}{6} E^3 \prod_{i=1}^2 \frac{1}{\sigma_i} \left(\frac{1}{B_i} \right)^{1/2} \prod_{j=1}^3 \frac{1}{\sigma_j B_j} \quad (9)$$

The density of states $\varrho^{\ddagger}(E)$ and $\varrho(E)$ (for the evaluation of $Q_{\text{rot}}^{\ddagger}Q_{\text{a}}$) as required in eq 6 can be obtained by convolution of the rotational density of states of eq 9 with the vibrational density of states (obtained by direct count of vibrational states^{10,11}) by using the convolution theorem² for separable vibrational and rotational Hamiltonians.

The implementation of eq 6 with eq 9 (which can easily be carried out with an RRKM program²) with the transition state chosen at the centrifugal barrier gives rise to the original Gorin result. This approach is known to produce a recombination rate coefficient that overestimates experimental results and an increase¹² in the rate coefficient in approximate proportion to $T^{1/6}$, in contrast to, for example, the recombination of two methyl radicals, which, according to most data, appears to have a slight negative temperature dependence.¹² The main cause¹³ of these failings is the use of a single transition state for all energies for any particular temperature, rather than variational selection of the transition state for each energy. Further, because the fragment interactions are treated as independent of the fragment orientation, the Gorin model neglects the effect of the geometry of the fragments on the outcome of a radical–radical collision.

A variational treatment of simple-fission processes requires a specification of the properties of the potential surface with variation of the fragment separation. In particular, the fragment rotations in the angles θ , ϕ (see Figure 3) undergo the greatest changes with progress toward reaction, and so an improved treatment of simple-fission reactions must involve a description

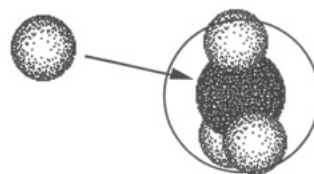


Figure 4. $\text{CH}_4 \rightarrow \text{CH}_3 + \text{H}$. The circle as drawn represents the sphere of radius r_{crit} . r_{crit} is the fragment separation at the onset of hindrance of the internal rotations of the CH_3 and H fragments.

of the potential function of interaction of the fragments. For any r^+ , the angular dependence of this function shall be approximated as a function in θ alone;³ we designate this potential function $V_{\text{int}}(r^+, \theta)$.

The modified Gorin model is the modification to Gorin's initial ideas⁹ provided by Benson.¹⁴ Benson added fragment interactions to the transition state potential function $V_{\text{int}}(r^+, \theta)$ by constraining fragment orientations with hard-sphere repulsion at contact of their constituent atoms. For increasing fragment separation the rotational freedom of the fragments will increase to the point where these rotations are unhindered; beyond a separation r_{crit} , the moieties are free to rotate. In the example shown in Figure 4 of the reaction $\text{CH}_4 \rightarrow \text{CH}_3 + \text{H}$, there is a critical separation r_{crit} of the centers of mass of the fragments beyond which the CH_3 radical is a free rotor. Benson's extension of the potential function thus provides a full reactive potential energy surface for the progress from reactants to products, although this description is known to be inadequate for small fragment separations at which hindered rotors are poor descriptions of bending vibrations. The principal consequence of this choice of $V_{\text{int}}(r^+, \theta)$ is that for $r^+ \leq r_{\text{crit}}$ some orientations of the fragments are disallowed.

For separations $r > r_{\text{crit}}$, the interaction potential between the two fragments will be independent of their orientation, i.e. everywhere attractive, as in the original Gorin model. So orientations for which an A–B chemical bond cannot form do nevertheless offer an attractive potential well within which transition state theory will predict (for any potential function) that the system can be stabilized, due to the weak van der Waals interactions of the nonbonded atoms. This is correct for very low temperatures, or for ion/molecule reactions, but contradicts intuition for recombining free radicals; we would expect that collisions occurring in unfavorable orientations would spend little time in the shallow van der Waals well and thus would not be capable of the exploration of the $\text{A}\cdots\text{B}$ phase space required to produce an A–B chemical bond, resulting in rapid separation of the fragments. Thus, inclusion of fragment interactions is alone insufficient to improve an RRKM/TST treatment upon the Gorin result, or indeed for any potential function.¹

The resolution of this difficulty lies in the exclusion of such orientations from the reactant phase space, as illustrated in Figure 5. From the viewpoint of transition state theory, this exclusion is equivalent to the dynamical assumption that the formation of AB cannot proceed through collisions in unfavorable orientations, which involves the assumption that energy transfer between fragments is too inefficient to make the lifetime in the van der Waals well comparable to that required for reorientation. The important consequence of this assumption is to enable the construction of a (usually spherical) dividing surface $r = r^+ < r_{\text{crit}}$; this surface correctly divides phase space in two, since unfavorable orientations are not part of reactant phase space. Note that this dividing surface passes within the hard-sphere radii of the fragment's atoms. The effect of placing the dividing surface in regions of the potential function inaccessible to trajectories is that only trajectories involving

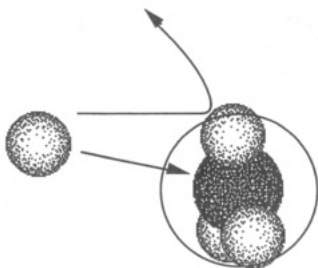


Figure 5. Restriction of reactant phase space. The circle as drawn represents the $r = r_{\text{crit}}$ spherical dividing surface which excludes trajectories with collisions in the incorrect orientation for reaction.

collisions in the correct orientation will produce AB. This dynamical assumption and the dividing surface necessary for the TST description of radical–radical recombination reactions have not previously been explicitly stated; it leads in this paper to a new understanding and improvement of the Gorin model, but should also be useful for more sophisticated potential functions.

For more sophisticated potential functions^{5,7,15} there is no well-defined onset of hindrance, but this dynamical assumption and dividing surface are implicit in almost all recent transition state theory calculations for radical–radical recombination reactions. An exception to this is the sinusoidally hindered rotor model (SHRM),³ which is discussed in the following section.

The density of states for a single hard-sphere hindered rotor can then be obtained by replacing the appropriate rotational constant in eq 9 with an effective rotational constant B_{eff} given by

$$B_{\text{eff}} = \frac{2}{1 - \cos \theta_{\text{hin}}} B_{\text{free}} \quad (10)$$

with evaluation of the rate coefficient then proceeding as in section II, with the important difference that a variational calculation is now possible. θ_{hin} is the average hindrance angle; that is, it is half the total angle of rotation possible for one fragment in the coordinate θ (see Figure 3). Thus, for ethane $0 \leq \theta_{\text{hin}} \leq 90^\circ$. B_{free} is the rotational constant of the corresponding two-dimensional free rotor. Implementation of the modified Gorin model requires evaluation of the hindrance angle θ_{hin} of the two hindered rotors at the transition state for any particular choice of $r^\ddagger$. Equation 10 gives rise to values of $B_{\text{eff}} > 2B_{\text{free}}$, which corresponds to the appropriate decrease in density of states with tightening of the fragment rotations toward vibrations. The hindrance properties of these rotors are functions of both angles θ and ϕ , but it is usual for convenience to specify θ_{hin} as a value averaged over ϕ .³ The calculation of θ_{hin} can be easily carried out, for example using the UNIMOL package.² Thus, the modified Gorin model can be implemented with minimal effort using standard programs; by evaluating θ_{hin} for a range of $r^\ddagger$, both microcanonical and canonical variational calculations can be performed.

Typical results^{3,4} with the modified Gorin model arise as follows. The hindrance of the two-dimensional rotations will vary with the fragment separation, as will the effective potential $V(r)$ between the two fragments; the variation with r of these entropic and enthalpic effects will mean that the variational minimum transition state will be energy or temperature dependent. As the energy or temperature is decreased, the separation $r^\ddagger$ at the variation minimum will increase toward the value r_{crit} ; this variation is due to the anisotropy of both the potential and rotational effects. For still lower energies or temperatures, the spherical dividing surface with radius $r^\ddagger < r_{\text{crit}}$ will no longer correspond to the variational minimum; the surface of a sphere

with radius $r^\ddagger > r_{\text{crit}}$ becomes the variation minimum dividing surface. Thus, two regimes arise: energies (or temperatures) for which variational minima arise at $r^\ddagger \leq r_{\text{crit}}$ and energies where they occur at $r^\ddagger > r_{\text{crit}}$. For $r^\ddagger > r_{\text{crit}}$ there is no anisotropy of the potential, but the variational minimum still changes with energy because the location of the centrifugal barrier is energy dependent.

The modified Gorin model predicts a transition from hindered to free rotor transition states at low temperatures. This conclusion is in qualitative agreement with experiment, because at sufficiently low temperatures all collisions must lead to recombination. However, attempting to describe the high- and low-temperature regimes with a single transition state theory rate coefficient reduces the accuracy achieved in the high-temperature regime, because the transition to free rotations in the transition state at lower temperatures arises for the modified Gorin model at a much higher temperature than physically reasonable, resulting in a large overestimate of the recombination rate coefficient.^{1,3}

It is curious that variational minima for $r^\ddagger > r_{\text{crit}}$ should arise in the modified Gorin model, since they appear inconsistent with the exclusion of unfavorable orientations from reactant phase space; this result is reexamined in the following section. Note that in calculations with more sophisticated potential functions^{5,7,15} than used in the modified Gorin model, all variational transition states are found for the temperatures of interest to arise for separations considered hindered by the modified Gorin model, i.e. $r^\ddagger \leq r_{\text{crit}}$, with the modified Gorin variational minimum $r^\ddagger$ for any specific temperature thought to be too large.

In addition, hard-sphere interactions have a catastrophic effect on calculated rate coefficients as $\theta_{\text{hin}} \rightarrow 0$ for small fragment separations; a classical treatment of these modes is inappropriate for small fragment separations, because as $\theta_{\text{hin}} \rightarrow 0$, the classical $Q_{\text{rot}}^\ddagger \rightarrow 0$, which is nonphysical (it should of course approach unity). More seriously, the variational behavior of calculated modified Gorin model rate coefficients for $r^\ddagger$ near r_{crit} is often discontinuous,⁴ or otherwise nonexistent,^{3,16} making assignment of a variational minimum rate coefficient for some temperatures ambiguous; these problems are illustrated for ethane decomposition in section V. However, better potential functions must also exhibit variational minima for $r^\ddagger > r_{\text{crit}}$ (or whatever distance is selected to exclude nonreactive orientations) and produce a discontinuity in the calculated rate coefficient, at sufficiently low temperatures. The modified Gorin model problems are not caused by the discontinuity in the modified Gorin potential function, but will arise whenever the dynamical assumption and dividing surface necessary for a TST description are used. This can be seen by realizing that a transition from $r^\ddagger < r_{\text{crit}}$ variational minima to $r^\ddagger > r_{\text{crit}}$ variational minima necessitates a discontinuity in the rate coefficient, whatever the potential function. The particular problem of concern for the modified Gorin model is that these problems arise in the temperature regime of experimental interest.

The above points suggest the following improved treatment for gas phase recombinations: a more appropriate Hamiltonian for these radical–radical reactions would assume that there was no coupling of rotations to the reaction coordinate for $r \geq r_{\text{crit}}$, because it is this coupling that produces a transition to $r^\ddagger > r_{\text{crit}}$ variational minima. An analogous assumption has been found to be essential for the TST description of Langmuir kinetics.¹⁷ Canonical variational TST for this Hamiltonian would produce a regime at high temperatures at which the variational minimum dividing surface would arise for $r \leq r_{\text{crit}}$, i.e. produce results equivalent to previous modified Gorin calculations, but at lower temperatures the variational minimum would arise at $r = r_{\text{crit}}$.

This zero-coupling limit (rather than the strong coupling of these modes assumed in previous modified Gorin calculations) is suggested to be more appropriate at the temperatures of experimental interest. This limit is not valid at very low temperatures, for which collisions of any orientation can produce reaction; that is, coupling of free rotations to the reaction coordinate is sufficient to reorient the system into the correct orientation for reaction. It is felt, however, that it is more appropriate to describe these two experimental regimes separately, rather than in one treatment.

It is of particular interest to see if this treatment for higher temperatures can at least partially redress some of the acknowledged weaknesses of the modified Gorin model such as too large rate coefficients, discontinuities in variational calculations, and perhaps even incorrect temperature dependence. These problems for the modified Gorin model have also been approached in two other ways: (i) smoothing the transition of hindered to free rotors with an appropriate potential function, as implemented in the sinusoidally hindered rotor model (SHRM),³ and (ii) interpolation of the transition state density of states between reactant and product.¹⁸ These two approaches are more complex than the present method and do not lead to the improvements found in the modified Gorin model in section V.

IV. Derivation of an Improved Treatment

In the derivation presented here, the formula for the rate coefficient calculated for the $r = r_{\text{crit}}$ dividing surface for our new Hamiltonian is established; this expression must be compared to the values of k_{uni}^{∞} or $k(E, J)$ calculated for the hindered regime to obtain the true variational minimum for any temperature or (E, J) . Note that the rate coefficient calculated at $r = r_{\text{crit}}$ is expected to provide a good approximate description of the results obtained by the modified Gorin model in the hindered regime. If the modified Gorin potential function is too poor a description of the hindered regime, as we shall see is the case for $\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6$, then the description derived below is suggested as an excellent approximate alternative.

The derivation in this section closely follows that of Gilbert and Smith² for $k(E, J)$ for reactions without a chemical barrier. The important differences between the analysis presented here and this previous derivation are restriction of the treatment to those trajectories with *sufficient energy in the reaction coordinate alone to surmount the centrifugal barrier* and applying this restriction to a specific dividing surface, that at the critical distance r_{crit} . The derivation here is applicable only to a special case of the more general treatment of uncoupled modes of Smith *et al.*¹⁹ The present result is valid only for a Hamiltonian with the following properties: (i) there is a distance r_{crit} at which there is a transition between complete coupling of some modes of the system and complete *uncoupling* of those same modes and (ii) the Hamiltonian for the coupled modes for $r > r_{\text{crit}}$ is independent of the reaction coordinate, as is the case in the modified Gorin model. The effect of (ii) is that the density of states or partition function of the coupled modes are constant for $r > r_{\text{crit}}$ and also strictly greater than the same quantity evaluated at $r = r_{\text{crit}}$; thus, the variational minimum rate coefficient must arise at $r = r_{\text{crit}}$ for sufficiently low temperatures or energies. The mathematical procedures of this derivation are identical in form to those presented by Smith *et al.*¹⁹ An analogous expression has previously been derived in the context of barrierless Langmuir adsorption.¹⁷

For systems for which external rotations must be taken into account, the microscopic rate coefficient $k(E, J)$ is given² by the expression

$$k(E, J) = \frac{1}{hQ(E)} \int d\mathbf{p}_u \int d\mathbf{u} \int_0^{\infty} dp_r \frac{p_r}{m} \delta[H - (E + R - R^\dagger)] \times S[E + R - V_{\text{eff}}(r^*)] \quad (11)$$

where the Hamiltonian $H = H(\mathbf{p}_u, \mathbf{u}, p_r, r = r^\dagger)$. The step function $S[E + R - V_{\text{eff}}(r^*)]$ ensures that there is sufficient total energy in the active modes of the reacting system to surmount the centrifugal barrier, i.e. $E + R > V_{\text{eff}}(r^*)$. If instead we wish to calculate the rate coefficient for $r^\dagger = r_{\text{crit}}$ with the insistence that there is sufficient energy E_r in the reaction coordinate r at the transition state to ultimately surmount the centrifugal barrier, then this step function must be altered to $S[E_r + V_{\text{eff}}(r^\dagger) - V_{\text{eff}}(r^*)]$, i.e. insisting that $E_r > V_{\text{eff}}(r^*) - V_{\text{eff}}(r^\dagger)$, where $\Delta V_{\text{eff}} = V_{\text{eff}}(r^*) - V_{\text{eff}}(r^\dagger)$ is the energy difference between the dividing surface at r_{crit} and the centrifugal barrier.

The Hamiltonian is rewritten as

$$H(\mathbf{p}_u, \mathbf{u}, p_r, r = r^\dagger) = E_r + H_u(r^\dagger) + V(r^\dagger, \mathbf{u} = 0) \quad (12)$$

Inserting the rewritten Hamiltonian and altered step function into eq 11, changing the integral over p_r to one over E_r , and inverting the order of integration, the following expression for the integrals over $\mathbf{u}$, $\mathbf{p}_u$, and E_r is obtained:

$$k(E, J) = \frac{1}{hQ(E)} \int_0^{E+R-R^\dagger-V(r^\dagger)} dE_r \int d\mathbf{p}_u \int d\mathbf{u} \delta[H_u - (E + R - R^\dagger - V(r^\dagger) - E_r)] S[E_r - \Delta V_{\text{eff}}] \quad (13)$$

Then, by incorporating the step function into the limits of the integral over E_r and recognizing the other two integrals as the density of states $\rho^\dagger(E + R - R^\dagger - V(r^\dagger) - E_r)$, eq 14 for $k(E, J)$ is reached:

$$k(E, J) = \frac{\int_{\Delta V_{\text{eff}}}^{E+R-R^\dagger-V(r)} dE_r \rho^\dagger(E + R - R^\dagger - V(r^\dagger) - E_r)}{hQ(E)} \quad (14)$$

If we now change the variable integration from E_r to $E_+ = E + R - R^\dagger - V(r^\dagger) - E_r$ and use the relation

$$V_{\text{eff}}(r^\dagger) = V(r^\dagger) + R^\dagger \quad (15)$$

then the “standard” form for $k(E, J)$ is obtained:

$$k(E, J) = \frac{\int_0^{E+R-V_{\text{eff}}(r^*)} dE_+ \rho^\dagger(E_+)}{hQ(E)} \quad (16)$$

The more typical form for $k(E, J)$ has $V_{\text{eff}}(r^\dagger)$ instead of $V_{\text{eff}}(r^*)$ in the upper limit of integration, as in eq 5. Equation 16 provides the means to carry out a microcanonical variational calculation of $k(E, J)$; for such a calculation for any E and J the results of eq 16 must be compared to the results for $k(E, J)$ obtained in the hindered region. The variational minimum $k(E, J)$ so obtained could then be canonically averaged to find a more accurate value for k_{uni}^{∞} . This more exacting approach is not pursued further here, but rather canonical variational TST calculations are carried out on the canonical average² over E and J of eq 16:

$$k_{\text{uni}} = \frac{1}{hQ_{\text{rot}}Q_a} \int_0^{\infty} dR \times \frac{e^{-R/kT}}{B} \int_{E_0(R)}^{\infty} dE e^{-E/kT} \int_0^{E+R-V_{\text{eff}}(r^*)} dE_+ \rho^\dagger(E_+) \quad (17)$$

where $E_0(R) = V(r^*) - \Delta R^* = V_{\text{eff}}(r^*) - R$ is the critical energy of reaction as a function of R . Inverting the order of integration of E and E_+ ,² and evaluating the resulting inner integral over E , we obtain

$$k_{\text{uni}}^{\infty} = \frac{kT}{h Q_{\text{rot}} Q_a} \int_0^{\infty} dR \times \frac{e^{-R/kT}}{B} \int_0^{\infty} dE_+ e^{-E_+/kT} Q^{\ddagger}(E_+) e^{-(V(r^*)+R^*-R)/kT} \quad (18)$$

The integral over E_+ can be identified as the transition state partition function $Q_a^{\ddagger}$ evaluated at r_{crit} ; this quantity is independent of R and so can be taken outside the integral over R . This leaves the following expression for k_{uni}^{∞}

$$k_{\text{uni}}^{\infty} = \frac{kT}{h} \frac{Q_a^{\ddagger}}{Q_{\text{rot}} Q_a} \int_0^{\infty} \frac{dR}{B} e^{-V_{\text{eff}}(r^*)/kT} \quad (19)$$

$V_{\text{eff}}(r^*)$ is dependent on R , and so this is the simplest form for k_{uni}^{∞} obtainable. This expression has superficial similarities to an expression previously used as an approximate model for reactions without a chemical barrier, for instance as discussed by Forst.²⁰ The important differences between eq 19 and this approximate treatment are that the transition state partition function is here evaluated at r_{crit} , not at some average centrifugal barrier, and no assumption has been made here as to the independence of $Q^{\ddagger}(E_+)$ of $r^{\ddagger}$.

Equation 19 requires an average over rotational energies of the height of the centrifugal barrier. The numerical determination of the centrifugal barrier height can be routinely performed using eq 3 given $V(r)$ and the variation of the rotational constant B_{inac} for the external inactive rotation with fragment separation. The evaluation of the integral is also routine; the details of these procedures are discussed in the following section as exemplified for the decompositions of ethane and neopentane.

V. Results

V.1. The Decomposition of Ethane. The reaction $\text{C}_2\text{H}_6 \rightarrow \text{CH}_3 + \text{CH}_3$ has been extensively studied both theoretically and experimentally as an archetypal simple-fission process. We shall not attempt to review this literature here. In this section results obtained from a standard modified Gorin calculation are compared with the results of eq 19, as well as with experiment and more exacting theory. Wagner and Wardlaw²¹ used microcanonical variational transition state theory with an adjustable potential parameter describing the tightness of the transition state to match the negative temperature dependence of the experimental data. Darvesh *et al.*¹² used a potential surface fitted to *ab initio* data and Monte Carlo techniques in their use of microcanonical variational transition state theory and also closely reproduced the experimental temperature dependence. The experimental data are those cited by Darvesh *et al.*¹² For a recent discussion of results in the falloff regime, see Davidson *et al.*²²

The Modified Gorin Treatment. Geometries, vibrations, and rotations of the fragments at the transition state are assumed to be product-like and independent of their separation, i.e. the methyl fragments are assumed to be planar at all fragment separations ($r > 3.0$ Å). The fragment interactions are specified by hard-sphere repulsion at contact of the constituent atoms, where the C and H atoms are treated as hard spheres inside 1.72 and 1.27 Å, respectively.²³ These interactions hinder the rotations of the methyl groups for $r < r_{\text{crit}} = 3.8753$ Å, at which the hindrance angle (necessarily for these planar groups) is 90° .

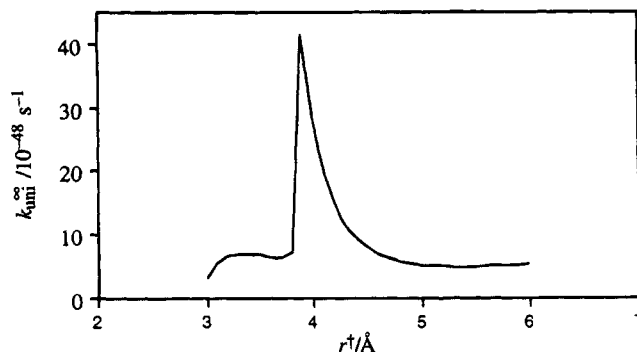


Figure 6. Variational behavior of the modified Gorin model for ethane at 300 K.

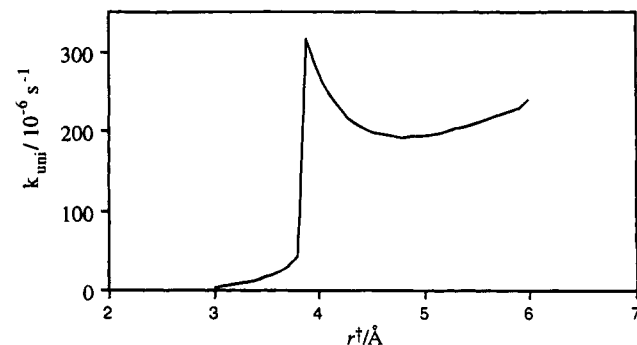


Figure 7. Variational behavior of the modified Gorin model for ethane at 900 K.

Programs RRKM and GEOM (from our public domain package UNIMOL²³) are used to evaluate the required parameters and rate coefficients, carrying out a canonical variational TST (CVTST) calculation for any particular temperature by finding the variational minimum rate coefficient k_{uni}^{∞} from the values calculated for $r^{\ddagger}$ in the interval $3.0 \text{ Å} \leq r^{\ddagger} \leq 6.0 \text{ Å}$ for temperatures in the range 300–900 K. The RRKM parameters held constant in this CVTST calculation are specified in Table 1.

Figures 6 and 7 show the variational behavior at 300 and 900 K, respectively. It is found for all temperatures examined, as illustrated in these figures, that no variational minimum is observed in the hindered region, i.e. for $r^{\ddagger} < 3.8753$ Å. It would be expected that a variational minimum would arise in the hindered region for the higher temperatures of the regime 300–900 K; however, the modified Gorin model treats the hindered rotations classically and thus permits the transition state hindered rotation partition functions to be less than unity, producing an unphysical description of the rate coefficient for these small values of $r^{\ddagger}$ for ethane. We are thus limited for ethane to selecting the modified Gorin variational minimum rate coefficient as that corresponding to the minimum in k_{uni}^{∞} for $r^{\ddagger} > r_{\text{crit}}$ for all of the examined temperatures: e.g. $k_{\text{uni}}^{\infty} = 4.86 \times 10^{-48} \text{ s}^{-1}$ at 300 K and $k_{\text{uni}}^{\infty} = 1.92 \times 10^{-4} \text{ s}^{-1}$ at 900 K. The discontinuities in the variational behavior in Figures 6 and 7 are not due to the hard-sphere interactions assumed in the modified Gorin model, but are actually caused by the dynamic assumption discussed in section III and the crossing of the reactant–product phase-space boundary by the variation calculation.¹⁷

Results of the Improved Treatment. Equation 19 is evaluated for the dividing surface $r^{\ddagger} = r_{\text{crit}}$. $Q_a^{\ddagger}$ can thus be evaluated assuming a separable Hamiltonian as

$$Q_a^{\ddagger} = Q_{\text{trans}}(Q_{\text{vib}}(\text{CH}_3))^2 Q_{\text{act}} Q_{\text{tors}} Q_{\text{hin}}^2 \quad (20)$$

TABLE 1: Parameters Held Constant in the CVTST Calculation for C₂H₆

Geometries of the Reactant ³ and Transition State ²⁷	
C ₂ H ₆	CH ₃
R _{C–C} = 1.534 Å	R _{C–H} = 1.093 Å
R _{C–H} = 1.093 Å	
∠HCC = 109.47°	∠HCH = 120.0°
staggered conformation	
hard-sphere radii	C: 1.72 Å
	H: 1.27 Å
Frequencies and Rotational Constants ^a	
reactant: C ₂ H ₆ ¹³	transition state: two planar CH ₃ groups ¹³
2954 (1)	3044 (2)
2896 (1)	580 (2)
1388 (1)	3162 (4)
1379 (1)	1396 (4)
995 (1)	
289 (1)	C ₂ H ₆
2985 (2)	B _{inac} = 1.215 (2;2)
2969 (2)	B _{ac} = 2.625 (1;3)
1472 (2)	
1469 (2)	TS
1206 (2)	B _{tors} = 9.572 (1;3)
821 (2)	B _{ac} = 2.395 (1;3)
Specification of the Morse Curve for V(r)	
dissociation energy ^b	D _e = 87.6 kcal mol ^{–1}
harmonic C–C stretch	ω = 995 cm ^{–1}
for input at r = 4.5 Å	V(r) = 87.0 kcal mol ^{–1}
fitted value	β = 1.925 Å ^{–1}

^a All values are in cm^{–1}, degeneracies or rotational dimensions; symmetry numbers in parentheses. ^b We use this value for consistency with previous SHRM and modified Gorin calculations.³

where Q_{trans} is the translational partition function of ethane, $Q_{\text{vib}}(\text{CH}_3)$ is the vibrational partition function of the methyl radical, Q_{act} is the partition function for the one-dimensional external active rotation, Q_{tors} is the partition function for the internal torsional rotation of the transition state, and Q_{hin} is the partition function for the hindered rotations of the methyl radical. The partition function Q for a one-dimensional rotation (assumed unhindered) with symmetry number σ is

$$Q = \frac{\pi^{1/2} (kT)^{1/2}}{\sigma B} \quad (21)$$

and so B_{tors} and B_{ac} (both for rotations of symmetry number 3) of Table 1 can be used with eq 21 to evaluate Q_{tors} and Q_{act} , respectively. The partition function Q_{hin} for a two-dimensional hindered rotor is evaluated using eq 10. For the methyl radical, $B_{\text{free}} = 9.572 \text{ cm}^{-1}$ and θ_{hin} at $r^\ddagger = r_{\text{crit}}$ is 90°, so $B_{\text{eff}} = 2B_{\text{free}}$. Thus, $Q_{\text{a}}^\ddagger$ is very easily evaluated. The partition function of the reactant $Q_{\text{tot}}Q_{\text{a}}$ is trivially evaluated using the data of Table 1.

The integral over R of eq 19 remains to be calculated. B is the rotational constant of the two-dimensional inactive external rotation of the reactant and so is independent of R . $V_{\text{eff}}(r^*)$ is R -dependent, as given by eq 3. For consistency with the modified Gorin calculation discussed above, $V(r)$ is assumed to be a Morse function in r ; under these conditions $V_{\text{eff}}(r^*)$ cannot be found analytically, and thus eq 19 must be evaluated numerically. To generate $V_{\text{eff}}(r^*)$ for any particular R , in addition to $V(r)$ the variation of the two-dimensional external inactive rotational constant $B^\ddagger$ with the fragment separation is required. These data can easily be generated for ethane from the geometries of the methyl fragments given in Table 1 using

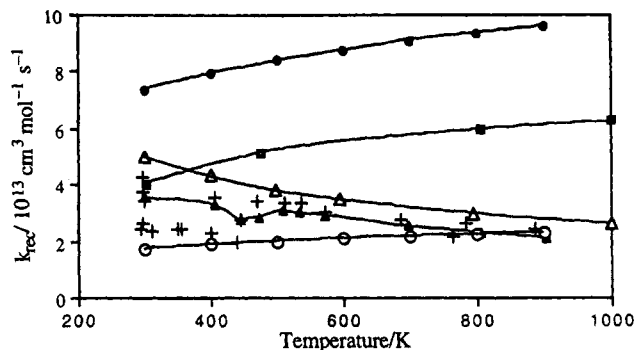


Figure 8. Temperature dependence of k_{rec} for the decomposition of ethane. Shown are eq 19 (open circles), experimental data (+),¹² SHRM³ (solid squares), modified Gorin (solid circles), and theoretical treatments (solid triangles²¹ and open triangles¹²).

the program GEOM. The values given assume that the methyl fragments are planar in the transition state; an improved treatment would account for the variation of the fragment geometry with r .

The results of eq 19 for temperatures in the range 300–900 K can be multiplied by the equilibrium constant to obtain k_{rec} , as plotted in Figure 8. It is clear from Figure 8 that eq 19 offers substantial improvements over the modified Gorin and SHRM models. These results still have a positive temperature dependence rather than the negative temperature dependence found for more sophisticated theoretical treatments; this may disappear with a microcanonical variational treatment. The temperature dependence has however flattened out and has best agreement with experiment at higher temperatures, underestimating k_{rec} at lower temperatures.

The SHRM provides an improved description of gas phase recombination processes in comparison to the modified Gorin model because of its sounder treatment of the potential function of the fragment rotations.³ This improvement reduces the calculated k_{rec} and variational minimum $r^\ddagger$ to be in closer accord with experiment than the modified Gorin model, but the variation minima obtained correspond to free rotational transition states and are thus expected to be overestimates of k_{rec} , according to the arguments expressed in this paper; this is borne out by Figure 8.

V.2. The Decomposition of Neopentane. The decomposition of neopentane to produce a *tert*-butyl radical and a methyl radical is a problematic system for theoretical treatment. The restriction of possible orientations at the transition state to reaction is large due to the bulky nature of the *tert*-butyl group, necessitating a description of this steric hindrance in any attempt to calculate the rate coefficient, but the size of the system has prevented its treatment with sophisticated potential functions thus far. Calculations that do not include steric hindrance are known to overestimate the experimental A-factor^{24,25} by several orders of magnitude.^{3,16} Unfortunately, however, calculations with the modified Gorin model fail to produce a variational minimum^{3,16} in the hindered regime and are excessively sensitive to the hard-sphere radii used to describe the constituent atoms.^{3,16} This problem can be avoided with the fitting of temperature dependent hindrance parameters and critical energies for a single transition state (for example Rao²⁴), but such methods offer little insight and no predictive ability. Thus, we turn to the methods discussed in the present paper as a possible solution to these difficulties.

For neopentane the parameters presented in Table 2 are used, with the methods discussed in the above section for ethane. For the geometries of the fragments given in Table 2, with hard-sphere radii of 1.72 and 1.27 Å for C and H, respectively, it is

TABLE 2: Parameters Used for the Decomposition of Neopentane

Geometries of the Reactant and Transition State			
C(CH ₃) ₄	C(CH ₃) ₃	CH ₃	
R _{C-C} = 1.537 Å	R _{C-C} = 1.504 Å	R _{C-H} = 1.079 Å	
R _{C-H} = 1.0843 Å	R _{C-H₁} = 1.086 Å	∠HCH = 120.0°	
	R _{C-H₂} = 1.092 Å		
∠HCC = 110.9°	∠H ₁ CH ₁ = 108.0°		
∠CCC = 109.47°	∠H ₁ CH ₂ = 107.2°		
∠HCH = 107.9°	∠HCC = 111.4°		
	∠CCC = 109.47°		
	pyramidal angle = 7.4°		
	out of plane bend = 27.7°		
Frequencies and Rotational Constants ^a			
C(CH ₃) ₄	C(CH ₃) ₃	CH ₃	C(CH ₃) ₄
2970 (2)	2971 (2)	3044 (2)	B _{inac} = 0.15828 (2;4)
2968 (3)	2966 (1)	580 (2)	B _{ac} = 0.14817 (1;3)
2966 (3)	2933 (3)	3162 (4)	
2860 (4)	2833 (3)	1396 (4)	
1480 (3)	1458 (3)		CH ₃
1462 (5)	1454 (2)		B _{methyl} = 9.572 (1;3)
1405 (1)	1453 (1)		
1377 (3)	1368 (2)		
1276 (3)	1330 (1)		C(CH ₃) ₃
1016 (2)	1261 (2)		B _{t-Bu} = 0.2599 (2;1)
972 (3)	1003 (1)		
901 (3)	925 (2)		
729 (1)	904 (3)		transition state
400 (3)	751 (1)		B _{tors} = 9.572 (1;3)
334 (2)	422 (2)		B _{ac} = 0.13969 (1;3)
210 (3)	158 (1)		
200 (1)	119 (2)		
	115 (1)		

Specification of the Morse Curve for $V(r)$
dissociation enthalpy²⁴ $\Delta H_0^0 = 83.1 \text{ kcal mol}^{-1}$
morse parameter³ $\beta = 1.875 \text{ \AA}^{-1}$

^a All values are in cm^{-1} , degeneracies or rotational dimensions; symmetry numbers in parentheses; frequencies from Rao and Skinner.²⁴

found (e.g., using UNIMOL²³) that r_{crit} is $4.0688 \text{ \AA}$, with θ_{hin} for the *tert*-butyl and methyl fragments 43° and 90° , respectively. $B_{free} = 0.2599 \text{ cm}^{-1}$ for *tert*-butyl. These hindrance angles (and thus k_{uni}^∞) are only weakly sensitive to the choice of the hard-sphere radii; a strong sensitivity of k_{uni}^∞ to these values has previously been reported, but does not affect the implementation of eq 19. Note that r_{crit} must be chosen to coincide with the onset of free rotations for the methyl fragment, rather than at the onset of free rotations for the *tert*-butyl fragment. k_{uni}^∞ is now calculated from eqs 19 and 20, producing the results given in Figure 9.

It is clear that eq 19 grossly overestimates k_{uni}^∞ for neopentane decomposition, with an error of 2 orders of magnitude for all temperatures in the range 700–1500 K. Recall that eq 19 is expected to be accurate only at low temperatures; this is dramatically shown in Figure 9. No experimental data are currently available for temperatures below 700 K, and so we cannot test its accuracy in this regime. Note that there is some uncertainty as to $V(r)$; both β and ΔH_0^0 are perhaps insufficiently well characterized for this reaction. However, the sensitivity to β is small in comparison to the discrepancy seen in Figure 9, and the slope (corresponding to ΔH_0^0) of the Arrhenius plots of the experimental and theoretical data are sufficiently similar to conclude that this uncertainty does not contribute to the difference between theoretical and experimental results for neopentane. Thus, a hindered rotor model of the transition state is inappropriate at these temperatures, and the variational minimum dividing surface must lie well inside r_{crit} .

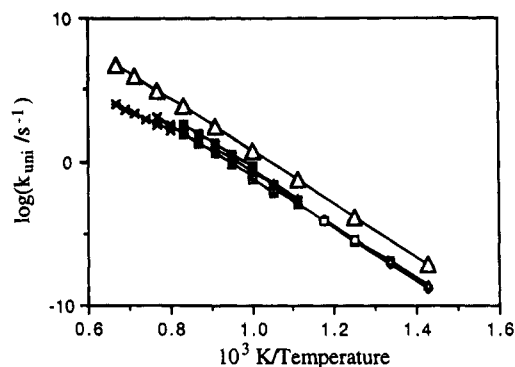


Figure 9. Decomposition of neopentane. Shown are results of eq 19 (open triangles) and experimental results (all others), as cited in Mitchell and Benson.²⁶

Hence, it is concluded that for bulky systems, such as neopentane, eq 19 is likely only to be useful at lower temperatures but can provide a useful upper bound to the decomposition and recombination rate coefficients.

For transition states in which the fragments are of unequal mass, a correction for the coupling of the motions of the two fragments may be necessary.¹⁶ It has been demonstrated¹⁶ however that this correction amounts to a reduction by a factor of only 2–3 in the calculated recombination and decomposition rate coefficients and so cannot account for the discrepancy observed in Figure 9.

This results provided by eq 19 are nevertheless better than previously obtainable with the modified Gorin or SHRM models;^{3,16} true variational minima are not obtained with these models for neopentane. Thus, it is not possible to extend the results of eq 19 to higher temperatures with the use of modified Gorin or SHRM approaches. Further work is clearly needed if the present treatment is to provide an adequate description of such strongly hindered systems. The importance of the present results is to demonstrate that hindered rotor models should not be used to model the transition states of bulky systems at high temperatures; this is made obvious by the present *predictive* calculations, whereas this cannot be concluded from previous modified Gorin model or empirical calculations. Note however that this does not rule out the use of eq 19 (which is after all strictly a low-temperature description) at lower temperatures; no low-temperature experimental results appear to be available for neopentane at the time of writing.

VI. Conclusions

The dynamical assumption and dividing surface necessary for the TST description of radical–radical recombination reactions have here been explicitly stated, with consequent improved understanding of the limitations of this description.

This paper has proposed an alternative Hamiltonian to that previously used in modified Gorin calculations of gas phase radical–radical recombination rate coefficients. This Hamiltonian assumes that there is no coupling between the reaction coordinate and other modes of the system outside the hindered regime specified by the modified Gorin potential function, rather than the strong coupling previously assumed. The principal effect of this alteration is that at low temperatures the variational minimum arises at the critical distance for the onset of hindrance, with no transition to free rotational transition states as $T \rightarrow 0$. An easily evaluated expression for the rate coefficient was obtained that produced improved results for both ethane and neopentane.

The modified Gorin model does not produce variational minima in the hindered region for ethane or neopentane, so there

is no intersection of the predictions of the variation of the rate coefficients with temperature for the modified Gorin model and eq 19, as there would be under more ideal circumstances. The modified Gorin model breaks down in having no variational minima (with discontinuities in the variational behavior of the rate coefficient) in the hindered region despite its implicit dynamic assumption, which is a logical inconsistency. This is a particular advantage of eq 19; it is not sensitive to these weaknesses of the potential function and can thus be used as a replacement of the modified Gorin model for description of the hindered regime for cases such as ethane, rather than just as an expression at lower temperatures. The rate coefficient calculated from a simple hard-sphere Gorin model is in general sensitive to the values used for the hard-sphere radii of the constituent atoms, whereas this is not the case for the present modification.

It thus can be concluded that the new Hamiltonian and the resulting expression for the rate coefficient given by eq 19 provide a useful tool for estimating the temperature dependence of recombination rate coefficients in conjunction with the modified Gorin model and are applicable to a wider range of systems than the modified Gorin and SHRM models. This method is best suited to smaller systems and lower temperatures, with increasing inaccuracy for bulky systems at higher temperatures. A particular advantage is that (unlike the simple Gorin models) the present description shows little sensitivity to the values of the assumed hard-sphere radii. The use of this method for bulky systems for low temperatures is an important future application.

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